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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

1- Semantics of Propositional Logic

1.1. Suppose you are a detective and know the following facts.

- If the knife is in the store room, then we saw it when we cleared the store room.
- The murder was committed at the basement or inside the apartment.
- If the murder was committed at the basement, then the knife is in the yellow dust bin.
- We did not see a knife when we cleared the store room.
- If the murder was committed outside the building, then we are unable to find the knife.
- If the murder was committed inside the apartment, then the knife is in the store room.

The question is: where is the knife?

First, we assigned symbols to the above clues:
s : the knife is in the store room
c : we saw the knife when we clear the store room
b : the murder was committed at the basement
a : murder was committed inside the apartment
y : the knife is in the yellow dust bin
o : the murder was committed outside the building
u : we are unable to find the knife

Formalize the facts as formulas and deduce where the knife is. Create an assignment of truth value for the propositional variables, and confirm that every fact has the truth value **true** for that assignment.

1.2. Let us consider a propositional language where

- A: “Angelo comes to the party”
- B: “Bruno comes to the party”
- C: “Carlo comes to the party”
- D: “Davide comes to the party”

Formalize the following sentences into a propositional logic formula.

(a) "Angelo comes to the party while Bruno doesn't"

(b) "Either Carlo comes to the party, or Bruno and Davide don't come"

(c) "If Angelo and Bruno come to the party, then Carlo comes provided Davide doesn't come"

(d) "Carlo comes to the party if Bruno and Angelo don't come, or if Davide comes"

(e) "If Angelo comes to the party then Bruno or Carlo come too, but if Angelo doesn't come to the party, then Carlo and Davide come"

1.3. For each of the following formulas draw its syntactic tree and indicate its subformulas. Remember the priority and associative conventions of logical operations given in the theoretical lecture.

(a) $p \wedge \neg q \to \neg p$;

(b) $p \wedge (\neg q \to \neg p)$;

(c) $\neg(s \to (\neg(p \to (q \vee \neg s))))$;

(d) $p \to (\neg q \vee (q \to p))$;

(e) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$.

1.4. Construct truth tables for each of the following propositions.

(a) $p \to p$

(b) $\neg q \wedge (p \to q)$

(c) $(p \wedge q) \to \neg(p \vee q)$

(d) $\neg(p \vee q) \to (\neg p \wedge \neg q)$

1.5. When $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$ determine if $\models_\nu \phi$, where ϕ is:

(a) $(p \vee q) \to (p \wedge q)$;

(b) $(p \vee \neg q) \wedge (\neg p \wedge q)$;

(c) $(p \to q) \wedge (p \to \neg q)$;

(d) $p \to (q \to (p \vee q))$;

(e) $((p \to q) \to p) \to p$;

1.6. Which of the following statements are valid? Justify.

(a) $(p \to q) \wedge (p \to r) \Leftrightarrow (p \to (q \wedge r))$

(b) $(p \to r) \wedge (q \to r) \Leftrightarrow ((p \wedge q) \to r)$

(c) $(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$

1.7. Without building the truth table, justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

(a) $\models p \to (((q \to p) \to p) \to r)$.

(b) $\models (p \to (q \to s)) \to (q \to (p \to s))$.

(c) $\models (p \to (q \to r)) \to ((p \to q) \to (p \to r))$.

(d) $\models ((p \wedge q) \to (r \vee s)) \to ((p \to r) \vee (q \to s))$.

1.8. Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

(a) If $\Gamma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Sigma \models \theta$.

(b) If $\Sigma \models \theta$ and $\Gamma \subseteq \Sigma$, then $\Gamma \models \theta$.

(c) If $\Gamma \models \theta$ and $\Sigma \models \theta$, then $\Gamma \cap \Sigma \models \theta$.

(d) If $\Gamma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cap \Gamma \models \theta$.

(e) If $\Sigma \models \theta$ then $\Sigma \cap \Gamma \models \theta$.

(f) If $\Sigma \models \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma \models \theta$.

1.9. For each of the truth functions *(a)*, *(b)* and *(c)*, construct a formula in disjunctive normal form and a formula in conjunctive normal form:

p	q	r	(a)	(b)	(c)
T	**T**	**T**	**T**	**F**	**T**
T	**T**	**F**	**T**	**T**	**F**
T	**F**	**T**	**T**	**F**	**T**
T	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**	**F**
F	**F**	**T**	**T**	**F**	**T**
F	**F**	**F**	**T**	**F**	**F**

1.10. Proving that sets of connective are complete.

(a) Show that the logical connectives $\{\neg, \vee\}$ are a complete set. That is, propositions of the form $p \to q$, $p \leftrightarrow q$, and $p \wedge q$ can be written using only the connectives $\neg$ and $\vee$.

(b) Define a new connective $\overline{\wedge}$ as follows:

p	q	$p \overline{\wedge} q$
T	**T**	**F**
T	**F**	**T**
F	**T**	**T**
F	**F**	**T**

Show that the propositions $\neg p$, $p \wedge q$, and $p \vee q$ can be written using only the connective $\overline{\wedge}$. Conclude that $\{\overline{\wedge}\}$ is a complete set of connectives.

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \to \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	